

# J-space, Part 2: The Model Matrix

From lab result to publishable finding — living handout

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Campaign started 2026-07-27 — preregistered before data (27c6d3c)

## 1 Where Part 1 left off, and what Part 2 must decide

Part 1 (Lab 37) replicated the descriptive geometry of the Anthropic workspace paper on `0lmo-3-32B-Think` at  $\sim 10\times$  thinner capacity, found the paper’s static causal dissociation *null* on both `OLMo` and `Qwen3.6-27B` under energy-matched instruments, established per-item frozen J-ablation as a control-clean cross-model causal handle on retrieved content ( $-2.9/-2.4$  nats), and measured a foil-calibrated 46-step workspace-ahead-of-text lead. Part 2 adjudicates *why the dissociation is missing*:

|    | hypothesis                                                       | discriminator                                                                                                                      |
|----|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| H1 | externalization (think-training moved the workspace into tokens) | Workstream A: matched-pretraining sibling <code>0lmo-3.1-32B-Instruct</code> , Qwen matrix completion, <code>gemma-4-31B-it</code> |
| H2 | occupancy (live J-space $\approx$ the output stream itself)      | Workstream D: output-occupancy index across the matrix                                                                             |
| H3 | training-lab specifics                                           | unfalsifiable; shrunk by exhausting the rest                                                                                       |
| H5 | residual instruments                                             | B (frozen-logit, fit-size, Dolma corpus, dose/persistence selection), C (load-demanding tasks, paper’s own sets)                   |

Standards:  $n \geq 60$  headline cells, two seeds on decisive grids, bootstrap CIs, BH-FDR across the matrix, preregistered directional predictions (`preregistration.md`).

## 2 Workstream A

### 2.1 A0 — does the J-dictionary survive post-training?

The part-1 Think lens, applied unchanged (donor  $J$ , recipient unembedding) to `0lmo-3.1-32B-Instruct` — same base model, different post-training. Preregistered gate: probes  $\geq 15/21$  and multihop pass@1  $\geq 0.85 \times$  donor = 0.2408.

**Verdict: TRANSFER\_FAIL, informative.** The readout basis is static across post-training (unembed row cosine median 0.9984; final-norm gain cosine 1.000), direct-fact probes transfer exactly (17/21 both), the shortlist transfers — what post-training blunted is specifically the *rank-1 sharpening*, the very component that was the J-lens’s distinctive advantage on Think (+41% relative pass@1 there, +8.5% here). The prereg FAIL rule fires: fit the sibling’s own 120-prompt lens. Recovery to  $\sim 0.28 \Rightarrow$  mid-band J-map drift under post-training; no recovery  $\Rightarrow$  Instruct resolves the silent bridge more weakly — itself an H1-flavored datum.

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\*Draft campaign handout; grows a subsection per banked phase. Part-1 handout: `jspace/handout/olmo32b_jspace_handout.pdf`.

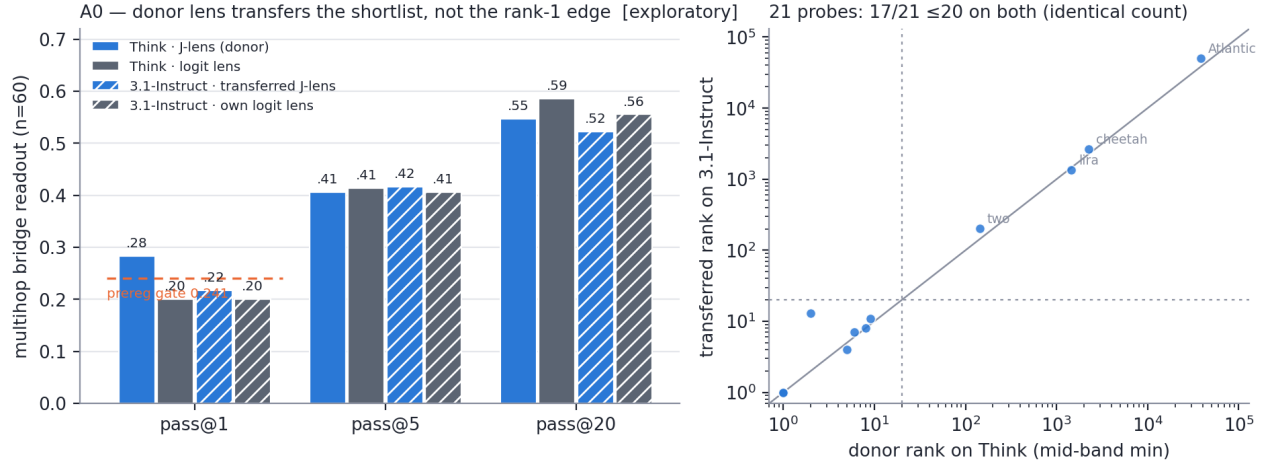


Figure 1: **A0 transfer gate.** Left: the transferred lens reads the bridge-entity *shortlist* at donor level (pass@5 0.417, pass@20 0.522 vs donor 0.41/0.55) but loses the rank-1 edge (0.217 vs 0.283, below the preregistered floor, dashed). Right: per-probe mid-band ranks — 17/21  $\leq 20$  on *both* models, same items missing.

### 3 Workstream B

#### 3.1 B3 — does the causal handle need the Jacobian? [exploratory pilot]

B3 — does the causal handle need the Jacobian? (OLMo-3-32B-Think, per-item frozen top-10, 95% boot CI) [exploratory pilot]

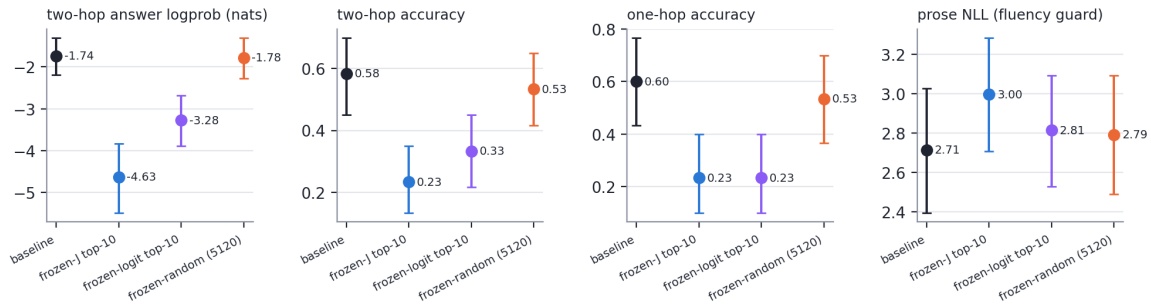


Figure 2: **B3.** The frozen-*logit* dictionary (no Jacobian, violet) reproduces  $\sim 53\%$  of frozen-J’s two-hop logprob deletion ( $-1.54$  vs  $-2.90$  nats) and the *identical* one-hop collapse ( $0.23=0.23$ ), while the random twin sits on baseline. Output-aligned directions alone carry much of the part-1 marquee effect; the geometry control family (R3) adjudicates the rest.

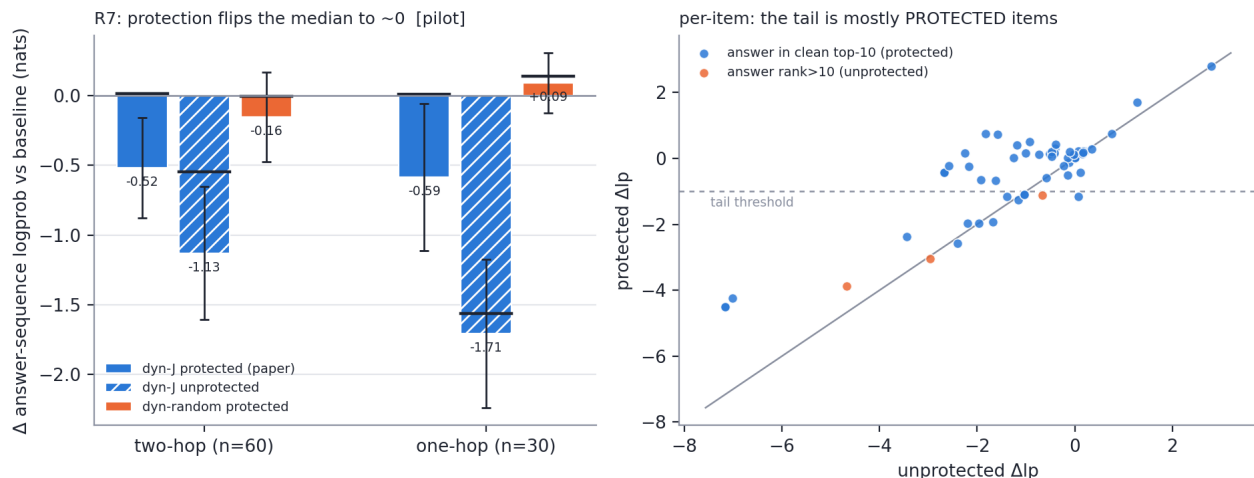


Figure 3: **R7 pilot (Think)**. Left: with the paper’s clean-output top-10 protection, the *median* item is untouched (two-hop +0.02, one-hop +0.01 nats) while the unprotected twin deletes hard (−0.55/−1.56 median) — part-1’s live-ablation damage was mostly output-stream deletion (H2, causal form). Right: a J-specific tail survives protection (16/60 two-hop items lose > 1 nat; protected-random on the same items: +0.34), and 13 of those 16 have the answer at clean rank 1–3 — *explicitly protected yet still deleted*: indirect J-content damage (bridge-entity reading), the confirmatory grid’s prime target.

## 4 Workstream R — assay repair (the addendum’s mandate)

### 4.1 R7 pilot — the paper’s output-protected ablation, run faithfully

### 4.2 R2 — capacity under the paper’s own estimator

### 4.3 R7 cross-model + R2-Qwen — the campaign headline (pilot)

### 4.4 The four-model ladder — the causal signature reorganizes

### 4.5 What the protected ablation actually deletes (tail mechanism)

The selected-id instrument logs, per layer and position, which dictionary rows the ablator deflates. On Olmo-3-32B-Think: the two-hop bridge entity’s tokens are deflated on **95%** of items versus **2%** under a bridge-reassignment permutation ( $p < 0.0005$ ) — the J-directions really do carry the intermediate entity. But this is *universal*: 0.94 on the damaged tail versus 0.95 on undamaged items, so bridge deletion alone is not sufficient for damage. What separates the tail is (i) protection failure — 21% of tail items had answers outside the protected clean-top-10 versus 0% of non-tail — and (ii) among the 19 tail items whose answer *was* protected, lower model confidence (median clean rank 2 versus 1). Deleting workspace content costs the model only when the answer is not already locked in.

### 4.6 Gemma-4-31B: an architecture datum with a depth

### 4.7 Design consequence: the confirmatory endpoint must change

The G6 power simulation, calibrated on the pilot parquets, finds that protected paired deltas are a zero-mode plus heavy-tail mixture. The drafted 0.5-nat *mean* primaries are structurally underpowered. This subsection’s original ICC figure ( $\approx 0.40$ , apparently uniform across models) is **superseded** — see §??.

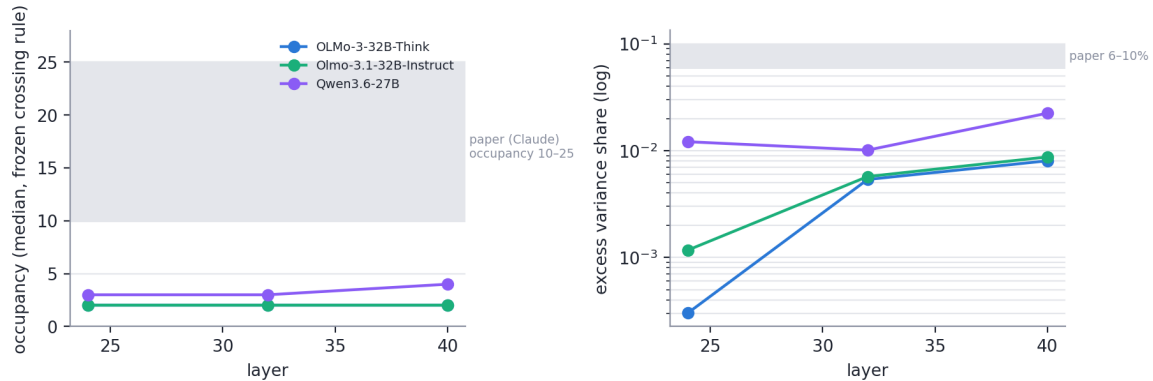


Figure 4: **R2 pilot**. Marginal-gain occupancy (frozen crossing rule, 3 random control dictionaries) and excess variance share at median occupancy. Both OLMo siblings: occupancy median 2, excess  $\leq 0.9\%$  — far below the paper’s Claude band (10–25, 6–10%), identical across the post-training pair. The Qwen recompute under this same estimator is the decisive cross-model capacity cell.

## 5 VM8 — the repair block that changed the design

A second forensic review was accepted and executed. Four repairs moved numbers that appear earlier in this handout; where they disagree, this section wins.

### 5.1 The clustering unit was wrong, and it was load-bearing

Every family-clustered interval, ICC and the power simulation above used `name.split("-")[0]` as the statistical family. That is a string accident: `atomic-80-state` and `ex-element-state-80-8` are the same template but landed in different families, sixteen unrelated items shared the family `ex` because their names begin with those letters, and each of the ten one-hop “capital of X” items was its own family. Under a hand-audited map (120 items, rule (*second-hop relation*, *answer type*)) the pilot’s 60 two-hop items are **25 canonical families, not 38 raw labels**, one of which holds 11 of the 60.

Point estimates are unchanged — clustering touches uncertainty, not means. What moved is the apparent homogeneity: audited ICC is 0.66 (base), 0.56 (Think), 0.17 (Instruct), 0.75 (Qwen), against a uniform 0.37–0.42 before. **One conclusion flips**: Instruct’s one-hop interval moves from  $-0.45[-1.03, +0.06]$  to  $-0.45[-1.29, -0.01]$ , so the “weak forward dissociation” rung of the four-model ladder is withdrawn — it reads as equal-depth damage. It is knife-edge and reverses again under family weighting, which is why the preregistration candidate fixes the weighting estimand in advance.

### 5.2 At 60 families, only two cells can carry a binary test

Inverting the power question — given  $m$  families, what is the smallest detectable effect at 90%? — gives, at the 60 canonical families the bank now holds: OLMo Think needs a 0.30 rate difference to see its 0.133 pilot gap ( $2.3\times$  short), Instruct 0.30 against 0.100 ( $3.0\times$  short). Only **Qwen on the rate endpoint** and **Instruct on the mean endpoint** are testable, and the two endpoints have *opposite* feasibility profiles across models. The candidate therefore states most hypotheses as estimation with intervals and declares a test only where power exists.

### 5.3 The averaged Jacobian’s in-band gap is a ceiling

For a *uniform* perturbation the per-position and corpus-averaged estimators are algebraically identical,  $\sum_s J_s d = P(\bar{J})d$ , so the earlier faithfulness test could not see position-averaging loss at all. Perturbing one position at a time — what the dynamic ablation does — separates them. Position-conditioning does **not** improve response

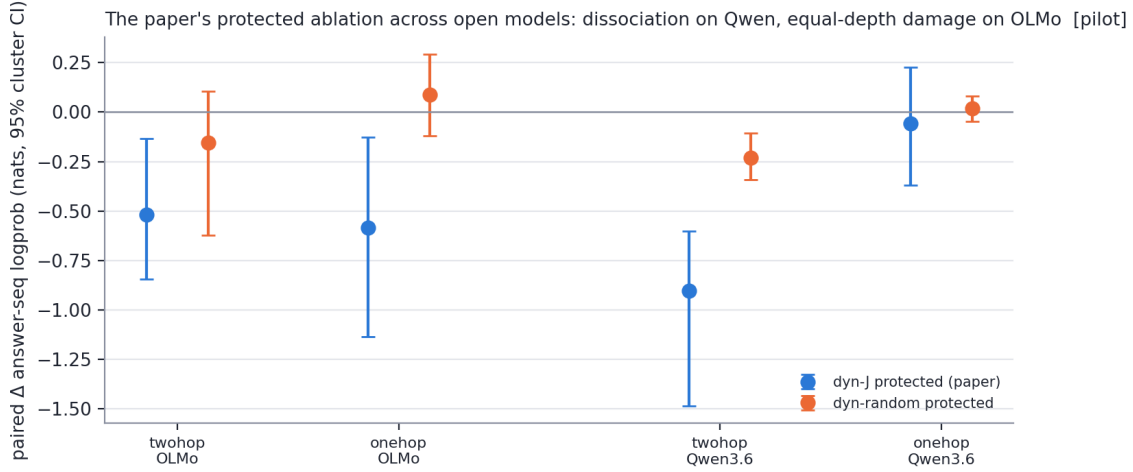


Figure 5: **Capacity and causal signature co-vary.** Under the identical protected protocol (paired family-clustered CIs): Qwen3.6-27B shows the paper’s dissociation shape — two-hop  $-0.91$   $[-1.49, -0.60]$ , one-hop  $-0.06$  [straddles 0] — while OLMo-3-32B-Think shows equal-depth content damage ( $-0.52/-0.59$ , both excl. 0, plus the indirect protected tail). The same models order the same way on paper-defined occupancy (3–4 vs 2, excess 1.0–2.2% vs  $\leq 0.9\%$ ), and BOTH sit an order of magnitude below the paper’s Claude band (10–25, 6–10%) — with lens fit size ruled out (Qwen used the published  $n=1000$  lens). Lens robustness: two independent 120-prompt Instruct fits agree at dictionary row-cos 0.993 and frozen-selection Jaccard 0.806.

direction at any band layer ( $-0.05, -0.04, -0.02$ ), though it roughly halves the magnitude error. Cell by cell, cosine tracks the ground truth’s *own* local linearity at within-layer Pearson  $r = 0.76-0.90$  ( $n = 144$ ): the estimator is not the binding constraint. Along the lens’s own top rows — where the ablation acts — in-band cosine is **0.58** versus **0.33** for random probes, so the often-quoted “ $\sim 0.2$  in band” overstates the problem for the actual intervention. The context-J methods arm fails its committed dev gate and is not admitted.

#### 5.4 Capacity: mislabelled, corrected, and still an order of magnitude low

The v1 estimator computed a centered activation and never used it; both shares were raw energy while the prose called them centered excess variance. Occupancy is unchanged (median 2 — it never depended on the share definition). Centered  $R^2$  excess is 0.49/1.02/**1.27%** against raw 0.03/0.54/0.80%. The headline survives in slightly stronger form.

#### 5.5 G5 passes, and catches the author

The task gate was run rather than waived. 1052 items, 15 excluded for answer-in-prompt leakage, zero bridge leakage, every two-hop item carrying its swap counterfactual, and **64 canonical families with  $\geq 3$  capable items** against a target of 60. The duplicate-fact audit caught nine newly authored “families” that re-expressed existing relations, four of them sharing literal facts — which would have let one fact land in both the confirmatory and the replication partition.

#### 5.6 Status

The preregistration CANDIDATE is written and the campaign is **stopped at the freeze boundary** awaiting principal-investigator sign-off. Five of eight gates pass; three conditions block, and all three need model weights absent from this machine: cross-model capability cohorts, a geometry-matched primary control, and lenses for both primary checkpoints. No confirmatory cell has run and the item partition has not been generated.

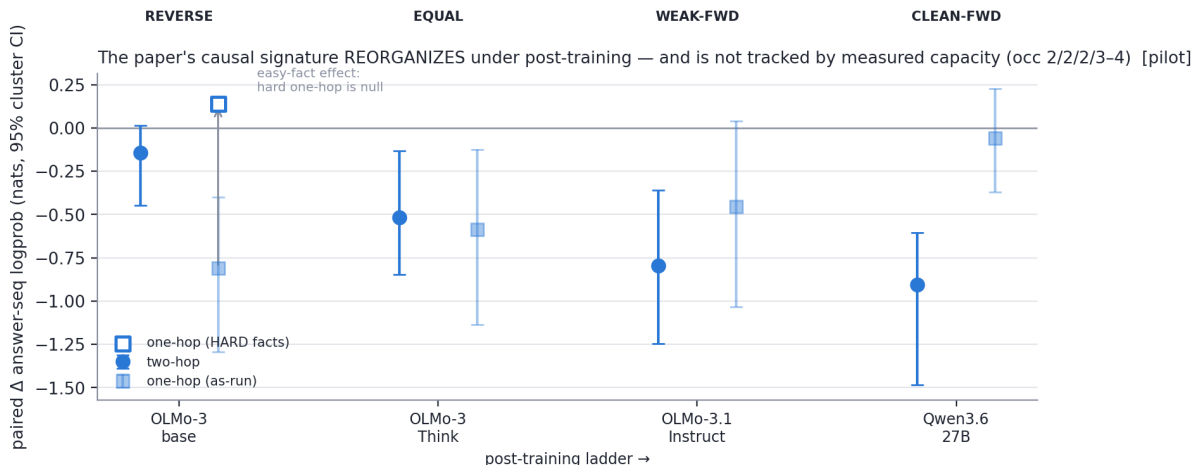


Figure 6: **The paper’s causal signature is not a fixed property of “an LLM” — it reorganizes along post-training.** Protected dynamic J-ablation, paired family-clustered CIs, four checkpoints: the base model shows a REVERSE dissociation (one-hop hit, two-hop spared), Think shows equal-depth damage, 3.1-Instruct a weak forward dissociation, Qwen the paper’s clean shape. Measured capacity does *not* track this (occupancy 2/2/2/3-4): Instruct and Think are identical on capacity and differ on shape. The open marker records the ceiling check that relabeled the bottom rung — on *hard* one-hop facts the base effect is null (+0.14; matched random -0.01), so the -0.81 was an EASY-FACT effect: rehearsed facts occupy the deletable output-adjacent channel and hard facts do not. Confirmatory hypotheses are therefore stated in fact-accessibility terms, not task depth.

## 6 VM9 — the freeze-blocker block: design decisions

The PI delegated the block’s open design questions. Three resolutions, each recorded with full rationale in `READY_FOR_FREEZE.md` (VM9 section) and preregistration-candidate §5:

### 6.1 The primary matched control

`dyn.energy.rank.matched.random`: per (item, layer, position), a random subspace matched *exactly, by construction* to the J arm’s achieved effective rank and removed-energy fraction at that site, orthogonal to the protected dictionary rows, deterministically seeded. For a span-projection intervention this is the *complete* geometric dose match — a subspace’s entire relation to the current state  $h$  is (rank, removed energy); there is exactly one principal angle between a subspace and a line — so the two arms differ only in *direction content*, which is precisely HP3’s specificity claim. `dynJ.rotated` was rejected as primary (rotated rows misalign with  $h$ ; selection scores and removed energy collapse toward the isotropic regime, re-introducing the dose confound) and `dyn.spectrum.matched.nonJ` likewise (a span projection is invariant to dictionary row spectrum given the span); both remain named secondary arms. Mechanical dev gates MC1–MC4 were committed before any run, with deliberately *no* behavioural gate, so the control cannot be tuned on outcomes.

### 6.2 The capability-cohort predicate

`capable.generation` — greedy decoding produces a frozen-alias answer — on each confirmatory checkpoint. Per-model difficulty windows are recorded but excluded from the predicate: window-based cohorts would re-select the bank per model, which is exactly the sampling bias the cohorts exist to close.

### 6.3 The capacity estimand (“does the data need recentering?”)

Yes: **globally centered  $R^2$**  is the confirmatory capacity estimand (`centered.variance.explained.excess`), with the raw energy share retained as named sensitivity and occupancy (crossing) unaffected — the  $v1 \rightarrow v2$

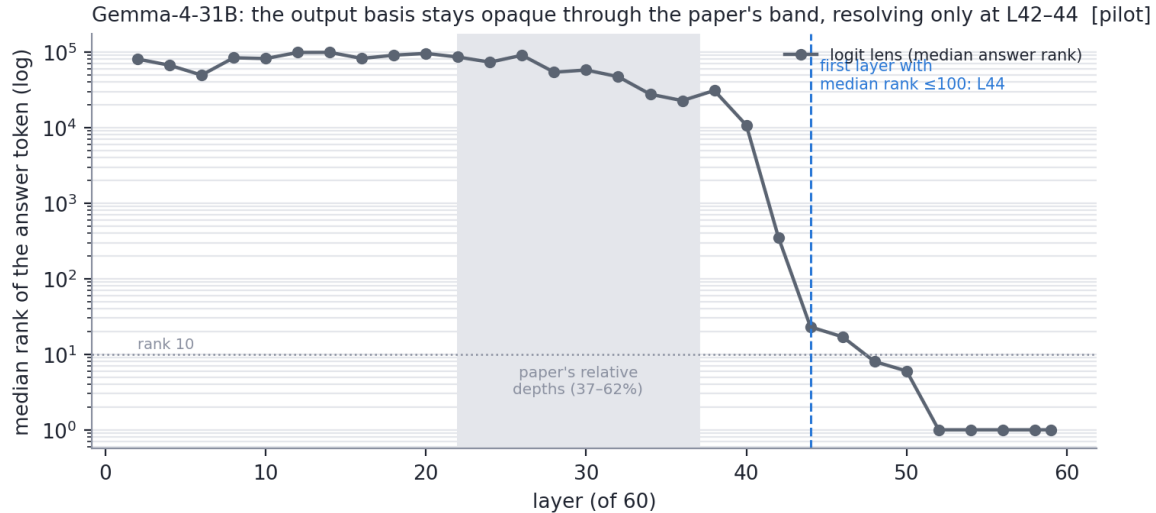


Figure 7: **Gemma’s residual stream is opaque in the output basis through exactly the band the paper studies.** Median rank of the answer token by layer, read with the reference unembedding (final norm, tied head, softcap 30). Ranks stay in the tens of thousands from L2 to L40 and collapse abruptly at L42–44, reaching rank 1 by L52. The paper’s relative depths (37–62%, i.e. L22–L37) sit entirely inside the opaque zone. The vanilla logit lens and the Jacobian micro-lens fail *equally* here, so this is a property of the model family, not of the instrument. Whether Jacobian transport rescues mid-band readability is the question the full 120-prompt fit answers; no family verdict is stated until it lands.

estimand repair already applied to Think, now extended to Qwen (published  $n=1000$  lens, pinned revision) and 3.1-Instruct (own lens, pilot revision).

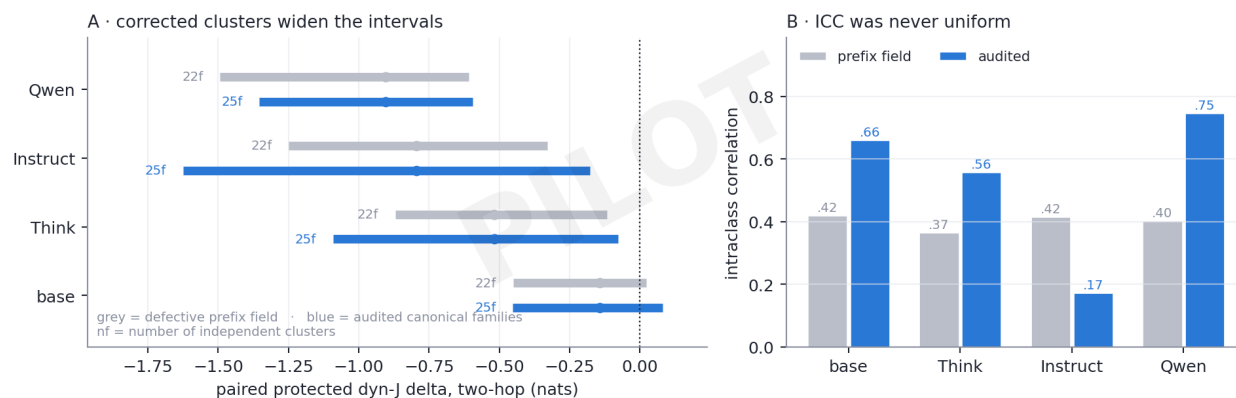
## 6.4 A ledger correction

“Neither primary lens exists” was wrong: the 3.1-Instruct lens has existed since the pilot as two independent registered fits (row-cos 0.993, selection Jaccard 0.806). Only the 3.1-*Think* lens was missing; its fit (recipe identical to the Instruct draw-A fit) runs this block.

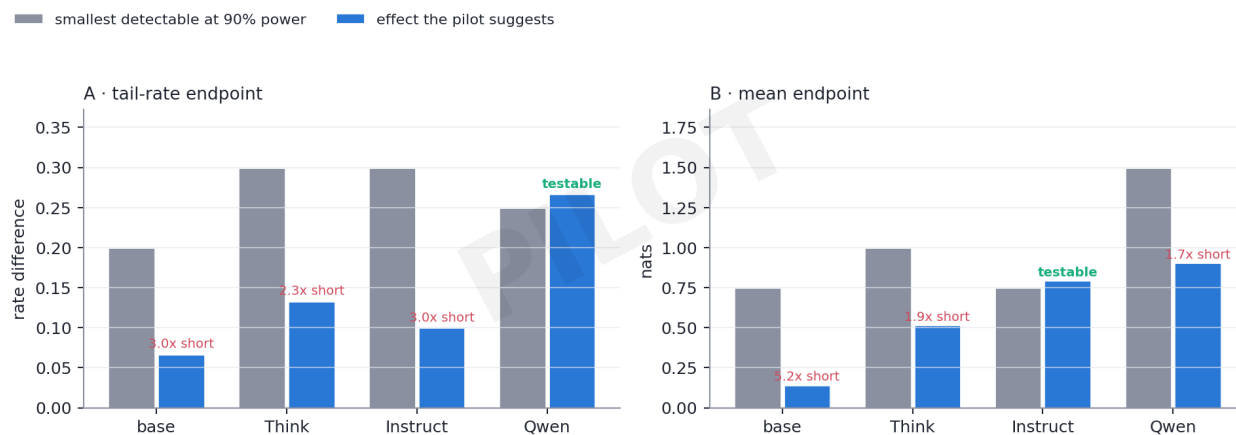
## Reproducibility

Run `dir special-lab-1/part2.20260727/` (metrics per model slug, lenses, logs); code + results mirrored to `interpretability/jspace/part2/` on branch `interp_jspace_part2`; every figure regenerates from metrics via `scripts/p2fig.py`; master dataset `report/matrix_master.csv`.

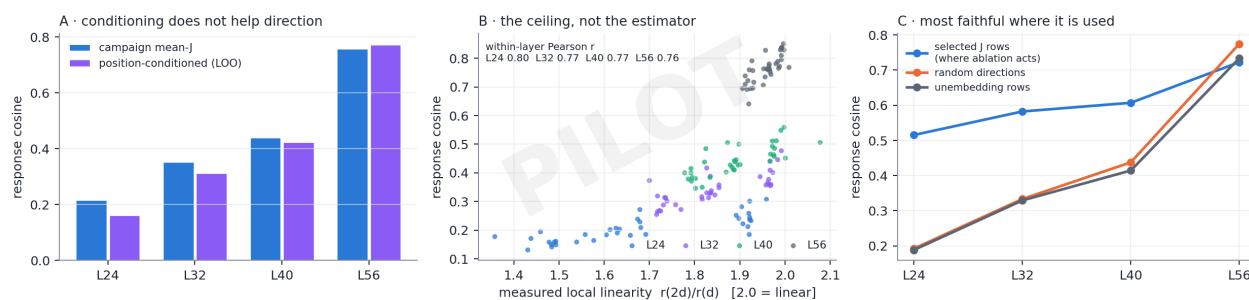
## The clustering unit was wrong, and it was load-bearing



## At 60 canonical families, only two cells can carry a binary test

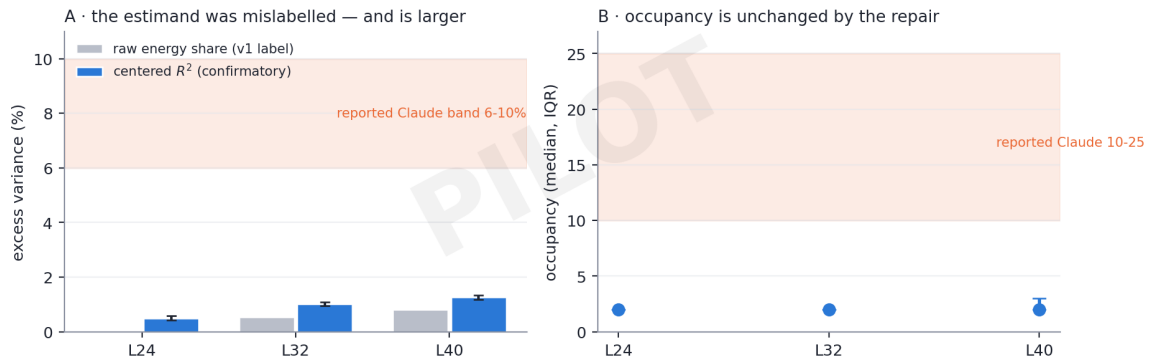


## D2 resolved: the averaged Jacobian's in-band gap is a linearity ceiling, not an estimation failure





### Corrected capacity on OLMo-3-32B-Think



### Stage-3 bank after the G5 audit

